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A Novel Blockchain-Based Model for Secure Genomic Data Management

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Abstract. Advancements in personalized medicine require secure, transparent, and privacy-preserving genomic data management systems. This study proposes a novel hybrid blockchain-based genomic data model integrating Self-Sovereign Identity (SSI), Decentralized Identifiers (DIDs), Verifiable Credentials (VCs), and decentralized storage Interplanetary File System (IPFS) to enable secure, privacy-compliant, patient-controlled genomic data exchange. The model leverages Polygon Proof-of-Stake (PoS) blockchain by deploying a smart contract that enforces fully access control functions applied on 100 samples of synthetic genomic data, ensuring only authorized researchers with valid DIDs and VCs can retrieve genomic data. Later, we performed five tests to evaluate our model performance. Security evaluations confirmed 100% data integrity validation through SHA-256 hash validation on-chain, ensuring tamper-proof genomic data storage. Unauthorized access attempts resulted in zero successful breaches, demonstrating the robustness of SSI-based authentication by showing revoked access. Additionally, IPFS data availability testing confirmed reliable and decentralized data retrieval through the Content Identifier (CID) on-chain. The model's access revocation mechanism enabled real-time patient control over genomic data access, ensuring compliance with GDPR privacy regulations. The proposed model provides a scalable, secure, and privacy-compliant solution for genomic data sharing in precision medicine, empowering patients with full control over their genetic data while facilitating researchers to trustworthy, decentralized data useability for precision research.

Keywords. Blockchain, Genomics, Personalized Medicine, SSI, DIDs.

1. Introduction

The rapid development of next-generation sequencing and genomics editing technologies have significantly transformed biomedical research and personalized medicine by enabling insightful genetic variations for disease mutations and evolutionary biology [1]. These high-throughput sequencing technologies further reduced the cost to approximately $600 per gene profile generated within few hours [2]. Gene sequencing has transitioned from an ancillary procedure to an essential component, becoming widely accessible in standard hospitals and clinics worldwide, rather than confined to specialized genomic research laboratories. However, Genomic data is widely used in precision medicine and oncology nowadays, identifying genetic mutations linked to

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chronic diseases i.e. cancer, diabetes, and rare disorders [3]. Therefore, their therapies are still not standard in general care due to the unavailability of gene data. Thus, the exchange of gene data is sensitive and fraught with challenges like data privacy, security, and ethical implications, which are crucial necessities in gene data management and its availability [4].

The traditional genomic data systems rely on centralized databases and cloud storage systems managed by intermediaries, i.e., research institutions, hospitals, or private companies. Data is typically stored in large repositories with strict access controls, requiring regulatory compliance, e.g., GDPR, making it inaccessible for research purposes. Blockchain technology solves many challenges in genomic data management, particularly security, data ownership, privacy preservation, access control, and ethical compliance. Also, earlier approaches often focus on one aspect, either consent or security, or centralization. By leveraging a democratic and tamper-proof nature, blockchain offers a hybrid solution to cope with the multiple challenges in practical applicability.

We developed a hybrid blockchain-based genomic data management model that contributes to three key research objectives to enhance data security, privacy, access control, and compliance in precision medicine. Firstly, we designed and implemented a model architecture to ensure secure access storage, privacy preservation, and consent control for synthetic genomic data by utilizing hybrid techniques, such as hashing, encryption, and DIDs to make genomic data available and useful to researchers. Secondly, deploying smart contract enforced access control policies to enable patients, researchers or institutions to securely register, grant, and revoke access to genomic data. Lastly, our model ensures only legitimate users can retrieve genomic records, preventing unauthorized access and misuse. The model performance was evaluated by testing gas cost, security, privacy, access control, and GDPR compliance for personalized medicine.

2. Related Work

Numerous approaches related to our proposed work explored from the literature to address the challenges mentioned above, and we observed the following categories:

- **Access Control and Consent Management:** Fine-grained access control mechanisms, such as Attribute-Based Encryption (ABE), ensure that only authorized users can access genomic data [5]. Dynamic consent models like MyData and GA4GH empower individuals to modify access permissions in real time, though challenges in usability and interoperability limits adoption [6].
- **Privacy-Preserving Techniques:** Homomorphic encryption (HE) enables computations on encrypted genomic data to ensure confidentiality. While Full Homomorphic encryption (FHE) offers high security with computational cost has led to more efficient alternatives like Partial Homomorphic encryption (PHE) [7].
- **Blockchain and Distributed Ledger Technologies:** Blockchain ensures secure and decentralized genomic data sharing through smart contracts and cryptographic access control [8]. Scalability and regulatory compliance remain challenges [5].

  Federated Learning for Genomic Data Analysis: Federated learning (FL) enables collaborative genomic analysis across institutions without direct data sharing [8] Despite its potential, communication overhead and model convergence require large-scale adoption optimization. Table 1 compares our model with related ones.

Table 1. Comparison of Proposed Model with Related Techniques

Ref.	Method	Security	Efficiency	Scalability	Privacy-Preservation	Challenges
[5,6]	(ABE)	Medium - Fine-grained access control	High - access control mechanisms	Medium- middle-scale genomic data	Medium- authorized access to specific data	Key Management, policy enforcement
[7]	(HE and FHE)	Medium- computations on encrypted data	Low- High computational cost	Low- Limited by computation overhead	High - Fully encrypted computations	Computational inefficiency, complexity
[8]	Blockchain-based Data Sharing	High - Immutable ledger ensures data integrity	Medium -large gene datasets	Medium- gene datasets	High- Transactions transparent	Scalability, regulatory compliance
[9]	Federated Learning (FL)	Medium- Model shared instead of raw data	High - Efficient decentralized data	High- deployed across multiple institutions	Medium- Data remains local	Communication overhead
Proposed Model	**Hybrid-Encryption, SSI, VCs**	**High- Hash validated Data integrity**	**High–IPFS availability, compliant GDPR.**	**High–large scale gene data**	**High- Authorized DIDs Access**	**Real Gene Data Transformation**

3. Proposed System Model

This section introduces the components and operational flow of our model architecture having entities and connections. Also, the hybrid blockchain-based techniques implemented by our model functioning as:

a) **Secure Hashing Algorithm (SHA-256):** This algorithm ensures data integrity by generating a unique 256-bit cryptographic hash for each genomic data record before storage. It validates retrieved genomic records against on-chain stored hashes.

b) **Advanced Encryption Standard (AES-256) Algorithm:** This algorithm encrypts genomic data before storing on IPFS to ensure privacy and security. It uses a symmetric key mechanism for efficient encryption and decryption.

c) **Role-Based Access Control (RBAC) Algorithm:** Implements on-chain access control using a smart contract, enabling real-time registration, grant, and revoke gene data access. Defines the hierarchical roles of patients, researchers, and VCs.

d) **Decentralized Identifier (DIDs) Authentication Algorithm:** This algorithm maps DIDs for patients and researchers to verify individual identities. It verifies VCs issued by trusted institutions before granting access.

The hybrid blockchain techniques enable an operational flow of genomic data for six key entities, including patients, researchers, VCs, i.e., healthcare institutions, blockchain network (Polygon PoS), smart contract (access-control layer), and off-chain storage (IPFS). Detailed working model architecture is illustrated in Figure 1. The platform accommodates two types of core users as previously defined, and the rest are system entities. Each user utilizes a DID generated by SSI framework, which has a public and private key combination by digital wallet (metamask). Patients are register themselves, assigned with their DIDs, and have full control over their synthetic genomic data to encrypt, granting or revoking access to researchers through smart contract enforced permissions. Researchers must authenticate using their DIDs and VCs issued by trusted healthcare institutions before requesting access to genomic data. The Polygon PoS, network stores SHA-256 hashes for tamper-proof data integrity, metadata, and

DIDs on-chain, while the actual genomic data record is securely stored off-chain in IPFS to minimize gas costs and ensure high scalability.

We developed a secure access process to connect patients, researchers, and institutions to interact with and operate our model. First, patients connect their wallet addresses through MetaMask, a prerequisite to use any blockchain application, register their DID, encrypt and upload genomic data to IPFS, and manage access permissions via our user-friendly web interface. Then, researchers can request access to data by authenticating using VCs, which is verified through smart contract. Once approved by SHA-256 hash, DIDs and VCs validation on-chain, the researcher views the IPFS data.

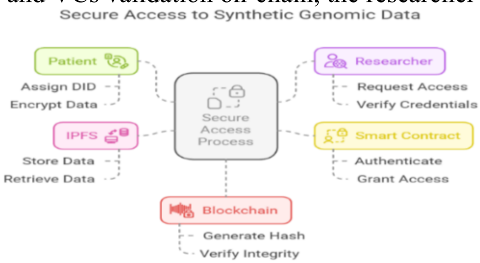

Figure 1. Proposed System Model

4. Material and Methods

This section elaborates our generated material, the implementation methods, and benchmarking of evaluation metrics to synthesize the results obtained by testing our model. To securely mimic real-world genomic data, 100 samples from 0-99 synthetic gene profiles were generated using the rule-based probabilistic method, upholding the structure like real data records to ensure GDPR compliance. Then, records were mapped to create SHA-256 hashes, Decentralized Identifiers (DIDs), and Metamask wallet addresses for on-chain transactions and authentication through our smart contract as mentioned in above section of system model. Our implementation method begins with setting up Polygon PoS Testnet and deploying solidity programmed smart contract via remix IDE as injected web. Then 100 SHA-256 hashes using Python library hashlib, 100 DIDs and metamask wallet addresses were mapped for authentication and transactions using a JavaScript library (Ether.js). AES encryption technique was applied before genomic data was uploaded to IPFS with retrieval using Web3.js library. Finally, using IPFS web service a node and its CID is created to make data available 24/7. We developed a frontend web interface using React to enable interactions between the entities to backend through and the Application Binary Interface (ABI) file which talks to the smart contract. Following this method and our model's architecture flow illustrated (Figure-1), we performed transactions on all samples and imported data in a CSV file from polygon scan logs. Using chart.js, we visualized our model's performance for i) Consumed gas for cost transactions of each function of SC ii) Validating SHA-256 hash integrity of retrieved genomic data to ensure enhanced security. iii) Simulating unauthorized access attempts ensuring authorized DIDs can access data to preserve privacy. iv) Testing data availability across IPFS node to assess decentralization and access control. v) Analyzing GDPR compliance through its three complaint components, i.e., Right to be forgotten, etc. Data and source code files are available as Github project: https://github.com/mutsh/Blockchain

5. Results

The implementation of our model successfully demonstrated test results for low gas cost, security, privacy-preservation, decentralized access control, and GDPR complaints for synthetic genomic data. Performance benchmarks showed low gas fees incurred by each SC function on Polygon PoS (Table 2), making the system cost-effective for large-scale genomic data storage and access management. The SHA-256 hash integrity verification test confirmed 100% uniqueness of retrieved genomic records matched until last byte over on-chain hashes, ensuring tamper-proof data storage as shown in Figure 2(A). Similarly, testing decentralized high data availability across IPFS node with retrieval time averaging 3 seconds per request is shown positive and accessible as in Figure 2(B) However, unauthorized access attempts resulted in zero successful breaches, validating DIDs effectiveness, and VCs authentication shown in Figure 3(A). Also, Figure 3(B) shows compliance of one of the main components of GDPR, i.e., "Right to be Forgotten," demonstrating that revoked users no longer have access to sensitive gene data being optimal solution to be applied in a real scenario.

Table 2. Description of Smart Contract Functions and Consumed Gas Fees Analysis

Function	Gas price	Description
Register DID ()	0.00220572369682191 POL	User can register their DID
Store Genomic Data ()	0.011518879137263058 POL	Store data, hash, DID on blockchain
Grant Access to Researcher ()	0.00168750446225893 POL	Patient can grant access to researcher
Revoke Access from Researcher ()	0.000986597862802515 POL	Patient can remove access to researcher
Check Access ()	N/A	Check access to DID records
Get IPFS()	N/A	Check permission to provide data link

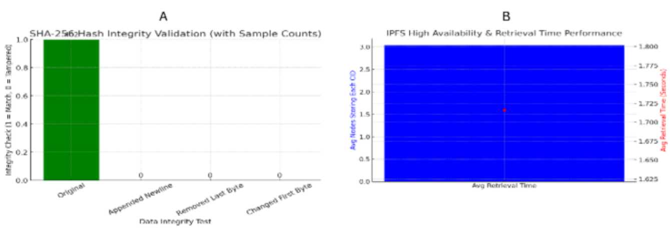

Figure 2. A: Data Hash Validation for Security, B: Access Control for Privacy- Preservation

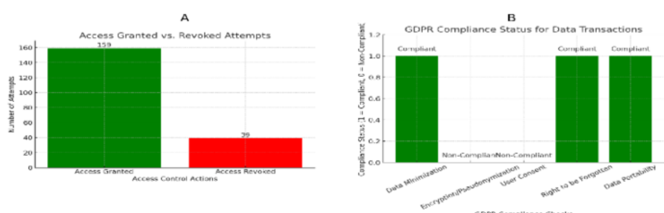

Figure 3. A: IPFS Node Data Retrieval Time, B: GDPR Complaint Components

6. Discussion

The high accuracy in SHA-256 hash validation confirms that blockchain-based storage mechanism prevents data tampering ensuring data authenticity. Moreover, the access control policies successfully prevented unauthorized access, highlighting the effectiveness of DIDs verification for preserving privacy. The rapid data retrieval speeds suggest IPFS has no data loss and is effective for genomic data availability. Tokenization ensured traceability and auditability, making data sharing secure and transparent. Compliance verification demonstrated that model adhering to legal privacy regulations. The current study leveraged 100 synthetic genomic data samples to simulate real genomic datasets while ensuring privacy and regulatory compliance to make it useful and accessible for research. Synthetic data allows for rigorous testing of the blockchain model without the risks associated with handling real sensitive patient data. The use of DIDs, VCs, and SC functions shows effective accessibility of genetic data to researchers to access it on request for precision therapies or tailored treatment plans. Since our model ensures data integrity, privacy preservation, and patient-controlled data sharing, patients can share access to their sensitive genomic data for monetary incentive. Despite the positive outcomes, challenges like gas fees on Polygon PoS, IPFS retrieval speed variations, and noncompliance of a few GDPR components were observed as methodological considerations. Future work will extend this model to enhance further advanced data security, compliance, and consent management processes on real genomic datasets using Machine Learning trials to validate this model's effectiveness in clinical applications for early disease prediction or verifiable drug sensitivity analysis.

7. Conclusions

Our model offers a novel solution for secure genomic data sharing with patient consent and access to data usability for research purposes, integrable with medical records. Leveraging blockchain's features, advanced cryptography, encryption, SSI, and access control, our findings demonstrate that this novel Blockchain-based model supports a scalable, secure, privacy-preserved, and efficient tool for personalized medicine clinical applications using genomic data.

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